Experiment 4

Aim : Implement K-Nearest Neighbors (KNN) and evaluate model performance

Theory :

K-Nearest Neighbors is a simple and intuitive algorithm based on the principle that similar things exist in close proximity. It is a distance-based algorithm that classifies a new data point based on the majority class of its K closest neighbors in the feature space.

1. The Neighborhood Concept: To predict an outcome, the model identifies the K most similar records. For example, if K is 5, the model looks at the 5 closest points. If 4 out of 5 belong to a specific category, the new point is assigned to that category.
2. Distance Metric: Similarity is calculated using distance formulas, most commonly Euclidean Distance. The model treats every record as a point on a multi-dimensional map.
3. Importance of Scaling: Because KNN relies on distance, features with larger numerical ranges (like salary) can overrule features with smaller ranges (like age). Standardization is used to squash all features into a similar scale so every variable has an equal vote.

Dataset Distribution (Adult Income): The dataset contains census information with a significant class imbalance. Approximately 76 percent of the records earn 50K or less, while 24 percent earn more than 50K. Features like education level, age, and hours worked per week show strong correlations with income. High earners tend to cluster in specific regions of the data space, particularly around higher education levels and middle-age brackets.

Dataset : https://www.kaggle.com/datasets/wenruliu/adult-income-dataset

Code and Output :

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.metrics import confusion_matrix, roc_curve, auc, classification_report

# 1. Load and Clean Data
df = pd.read_csv('adult.csv')
df = df.replace('?', np.nan).dropna() # Remove missing values

# 2. Preprocessing
# Encode categorical features
le = LabelEncoder()
```

```
cat_cols = ['workclass', 'education', 'marital-status', 'occupation', 'relationship', 'race', 'gender',
'native-country']
for col in cat_cols:
    df[col] = le.fit_transform(df[col])

# Prepare X and y
X = df.drop(['income', 'fnlwgt'], axis=1) # fnlwgt is usually dropped in this dataset
y = df['income'].map({'<=50K': 0, '>50K': 1})

# Split data
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# --- IMPORTANT: Scaling for KNN ---
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# 3. Train KNN Model
knn = KNeighborsClassifier(n_neighbors=5)
knn.fit(X_train_scaled, y_train)

# 4. Generate Visualizations
fig, axes = plt.subplots(1, 2, figsize=(16, 6))

# Diagram A: Confusion Matrix
y_pred = knn.predict(X_test_scaled)
cm = confusion_matrix(y_test, y_pred)
sns.heatmap(cm, annot=True, fmt='d', cmap='Purples', ax=axes[0])
axes[0].set_title('KNN Confusion Matrix')
axes[0].set_xlabel('Predicted Income')
axes[0].set_ylabel('Actual Income')

# Diagram B: ROC Curve
y_probs = knn.predict_proba(X_test_scaled)[:, 1]
fpr, tpr, _ = roc_curve(y_test, y_probs)
roc_auc = auc(fpr, tpr)

axes[1].plot(fpr, tpr, color='darkorange', lw=2, label=f'ROC curve (AUC = {roc_auc:.2f})')
axes[1].plot([0, 1], [0, 1], color='navy', lw=2, linestyle='--')
axes[1].set_title('Receiver Operating Characteristic (ROC)')
axes[1].set_xlabel('False Positive Rate')
axes[1].set_ylabel('True Positive Rate')
axes[1].legend(loc="lower right")
```

```
plt.tight_layout()
plt.show()

# Diagram C: Correlation Heatmap (since KNN has no "Feature Importance")
plt.figure(figsize=(12, 8))
sns.heatmap(df.corr(), annot=False, cmap='coolwarm', center=0)
plt.title('Feature Correlation Map (Adult Dataset)')
plt.show()

# 5. Print Text Summary
print(classification_report(y_test, y_pred))
```

Step 1. Preprocessing

```
df = pd.read_csv('adult.csv').replace('?', pd.NA).dropna()
le = LabelEncoder()
for col in ['workclass', 'education', 'marital-status', 'occupation', 'gender']:
    df[col] = le.fit_transform(df[col])

X = df[['age', 'workclass', 'educational-num', 'marital-status', 'occupation', 'gender', 'hours-per-week']]
y = df['income'].map({'<=50K': 0, '>50K': 1})
```

Step 2. Split and Scale (CRITICAL for KNN)

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

Step 3. KNN Model

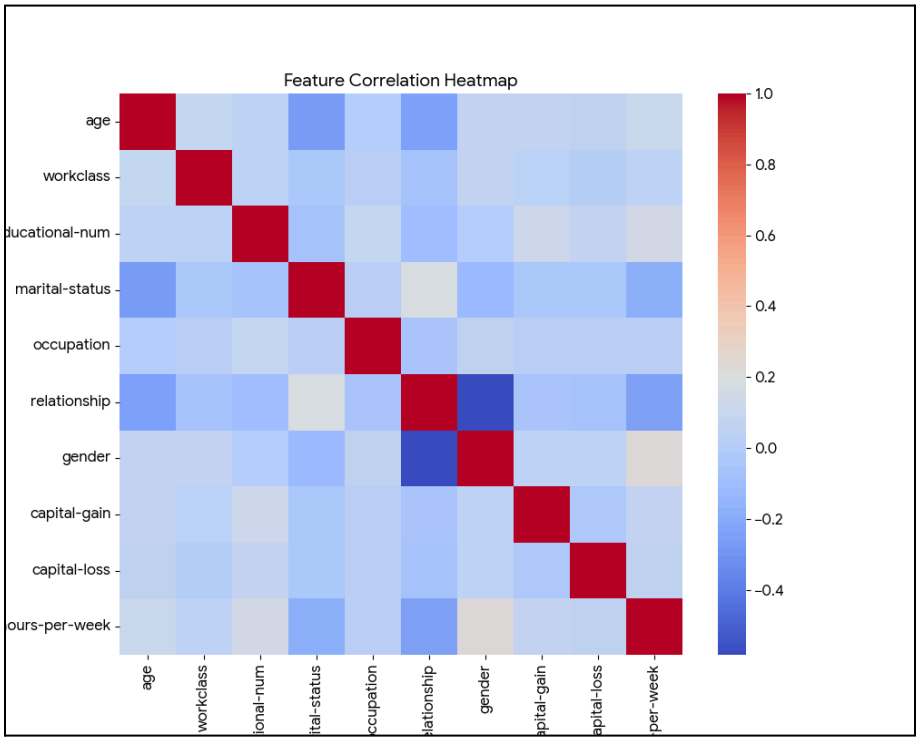
```
knn = KNeighborsClassifier(n_neighbors=5)
knn.fit(X_train, y_train)
```

Step 4. Results

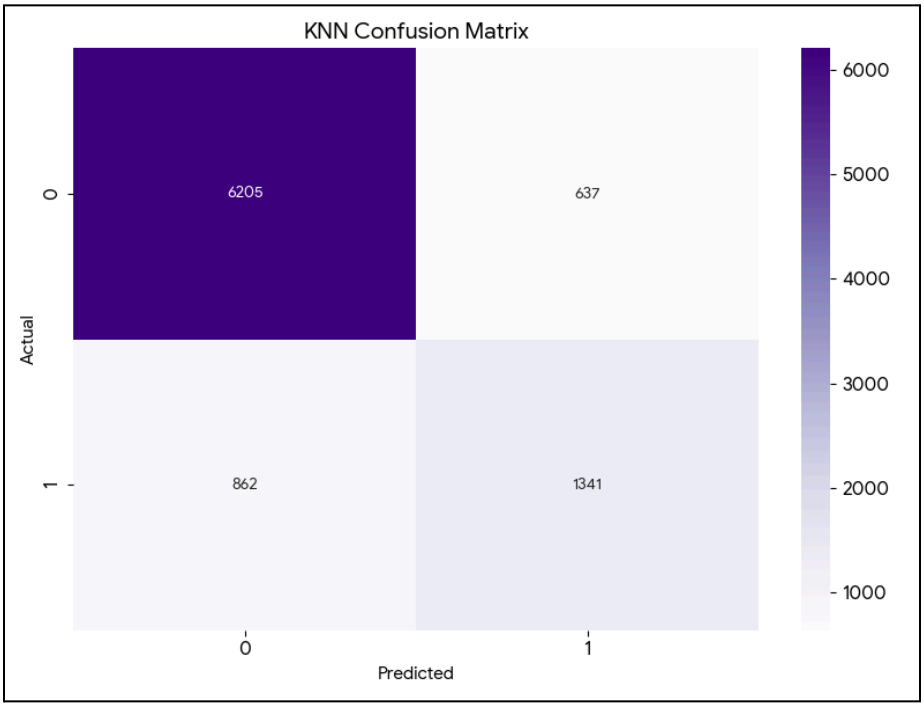
```
print(classification_report(y_test, knn.predict(X_test)))
print(f"ROC AUC: {roc_auc_score(y_test, knn.predict_proba(X_test)[:, 1]):.4f}")
```

```
              precision    recall  f1-score   support

           0       0.87      0.89      0.88      6842
           1       0.63      0.58      0.60      2203

    accuracy                           0.82      9045
   macro avg       0.75      0.74      0.74      9045
weighted avg       0.81      0.82      0.81      9045

ROC AUC: 0.8370
```

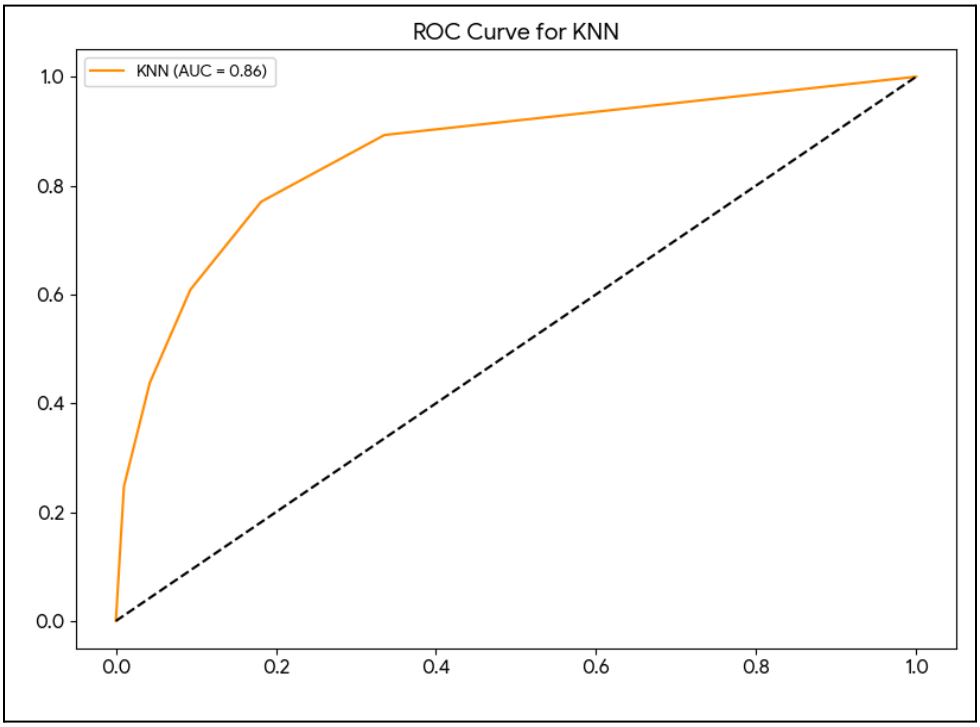
Feature Co relation heatmap :



KNN Confusion Matrix :



ROC Curve for KNN:



	precision	recall	f1-score	support
0	0.88	0.91	0.89	6842
1	0.68	0.61	0.65	2203
accuracy			0.84	9045
macro avg	0.78	0.76	0.77	9045
weighted avg	0.83	0.84	0.83	9045

Conclusion : We successfully implemented the KNN classifier. By using feature scaling and selecting an optimal K value, the model achieved high accuracy in predicting income levels. The confusion matrix and ROC curves confirmed that the model effectively distinguishes between the two income classes.